Blocking Colored Point Sets*

Greg Aloupis, Brad Ballinger, Sébastien Collette, Stefan Langerman, Attila Pór, and David R. Wood

Abstract This paper studies problems related to visibility among points in the plane. A point x *blocks* two points v and w if x is in the interior of the line segment $\overline{vw}$. A set of points P is *k-blocked* if each point in P is assigned one of k colors, such that distinct points $v, w \in P$ are assigned the same color if and only if some other point in P blocks v and w. The focus of this paper is the conjecture that each k-blocked set has bounded size (as a function of k). Results in the literature imply that every 2-blocked set has at most 3 points, and every 3-blocked set has at most 6 points. We prove that every 4-blocked set has at most 12 points, and that this

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G. Aloupis • S. Collette
Chargé de Recherches du F.R.S.-FNRS, Département d'Informatique,
Université Libre de Bruxelles, Brussels, Belgium
e-mail: aloupis.greg@gmail.com; sebastien.collette@ulb.ac.be

B. Ballinger
Department of Mathematics, Humboldt State University, Arcata, CA, USA
e-mail: bjb86@humboldt.edu

S. Langerman
Maître de Recherches du F.R.S.-FNRS, Département d'Informatique,
Université Libre de Bruxelles, Brussels, Belgium
e-mail: stefan.langerman@ulb.ac.be

A. Pór
Department of Mathematics, Western Kentucky University, Bowling Green, KY, USA
e-mail: attila.por@wku.edu

D.R. Wood (✉)
Department of Mathematics and Statistics, The University of Melbourne, Melbourne, Australia
e-mail: woodd@unimelb.edu.au.

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bound is tight. In fact, we characterize all sets $\{n_1, n_2, n_3, n_4\}$ such that some 4-blocked set has exactly n_i points in the ith color class. Among other results, for infinitely many values of k, we construct k-blocked sets with $k^{1.79\cdots}$ points.

1 Introduction

This paper studies problems related to visibility and blocking in sets of colored points in the plane. A point x *blocks* two points v and w if x is in the interior of the line segment $\overline{vw}$. Let P be a finite set of points in the plane. Two points v and w are *visible* with respect to P if no point in P blocks v and w. The *visibility graph* of P has vertex set P, where two distinct points $v, w \in P$ are adjacent if and only if they are visible with respect to P. A point set B *blocks* P if $P \cap B = \emptyset$ and for all distinct $v, w \in P$ there is a point in B that blocks v and w. That is, no two points in P are visible with respect to $P \cup B$, or alternatively, P is an independent set in the visibility graph of $P \cup B$.

A set of points P is *k-blocked* if each point in P is assigned one of k colors, such that each pair of points $v, w \in P$ are visible with respect to P if and only if v and w are colored differently. Thus, v and w are assigned the same color if and only if some other point in P blocks v and w. A *k-set* is a multiset of k positive integers. For a k-set $\{n_1, \ldots, n_k\}$, we say P is $\{n_1, \ldots, n_k\}$-blocked if it is k-blocked, and for some labeling of the colors by the integers $[k] := \{1, 2, \ldots, k\}$, the ith color class has exactly n_i points, for each $i \in [k]$. Equivalently, P is $\{n_1, \ldots, n_k\}$-blocked if the visibility graph of P is the complete k-partite graph $K(n_1, \ldots, n_k)$. A k-set $\{n_1, \ldots, n_k\}$ is *representable* if there is an $\{n_1, \ldots, n_k\}$-blocked point set. See Fig. 1 for an example.

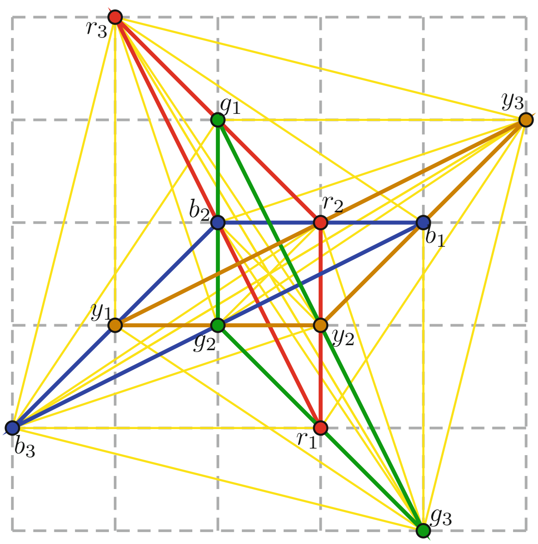


Fig. 1 A $\{3, 3, 3, 3\}$-blocked point set

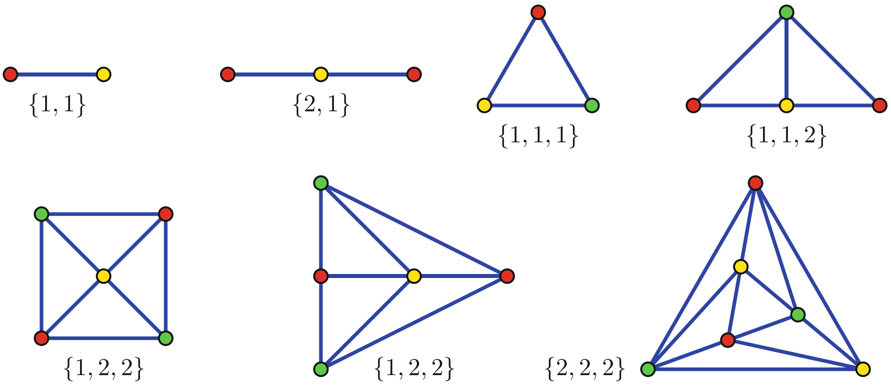


Fig. 2 The 2-blocked and 3-blocked point sets

The following fundamental conjecture regarding k-blocked point sets is the focus of this paper.

Conjecture 1. For each integer k, there is an integer n such that every k-blocked set has at most n points.

As illustrated in Fig. 2, the following theorem is a direct consequence of the characterization of 2- and 3-colorable visibility graphs by Kára et al. [5].

Theorem 2. *$\{1,1\}$ and $\{1,2\}$ are the only representable 2-sets, and $\{1,1,1\}$, $\{1,1,2\}$, $\{1,2,2\}$, and $\{2,2,2\}$ are the only representable 3-sets.*

In particular, every 2-blocked point set has at most three points, and every 3-blocked point set has at most six points. This proves Conjecture 1 for $k \leq 3$.

This paper makes the following contributions. Section 2 introduces some background motivation for the study of k-blocked point sets and observes that results in the literature prove Conjecture 1 for $k = 4$. Section 3 describes methods for constructing k-blocked sets from a given $(k-1)$-blocked set. These methods lead to a characterization of representable k-sets when each color class has at most three points. Section 4 studies the $k = 4$ case in more detail. In particular, we characterize the representable 4-sets, and conclude that the example in Fig. 1 is in fact the largest 4-blocked point set. Section 5 introduces a special class of k-blocked sets (so-called midpoint-blocked sets) that lead to a construction of the largest known k-blocked sets for infinitely many values of k.

Before continuing, we prove some basic properties about k-blocked point sets.

Lemma 3. *At most three points are collinear in every k-blocked point set.*

Proof. Suppose that four points p,q,r,s are collinear in this order. Thus, (p,q,r,s) is an induced path in the visibility graph. Thus, p is not adjacent to r and not adjacent to s. Thus, p, r, and s have the same color. This is a contradiction since r and s are adjacent. Thus, no four points are collinear. $\square$

A set of points P is in *general position* if no three points in P are collinear.

Lemma 4. *Each color class in a k-blocked point set is in general position.*

Proof. Suppose on the contrary that three points from a single color class are collinear. Then no other points are in the same line by Lemma 3. Thus, two of the three points are adjacent, which is a contradiction. □

Note that "convex position" in this paper means "strictly convex position."

2 Some Background Motivation

Much recent research on blockers began with the following conjecture by Kára et al. [5].

Conjecture 5 (Big-Line-Big-Clique Conjecture [5]). For all integers t and ℓ, there is an integer n such that for every finite set P of at least n points in the plane:

- P contains ℓ collinear points, or
- P contains t pairwise visible points (that is, the visibility graph of P contains a t-clique).

Conjecture 5 is true for $t \leq 5$ but is open for $t \geq 6$ or $\ell \geq 4$; see [1,9]. Given that, in general, Conjecture 5 is challenging, Jan Kára suggested the following weakening.

Conjecture 6 ([9]). For all integers t and ℓ, there is an integer n such that for every finite set P of at least n points in the plane:

- P contains ℓ collinear points, or
- The chromatic number of the visibility graph of P is at least t.

Conjecture 5 implies Conjecture 6 since every graph that contains a t-clique has chromatic number at least t.

Proposition 7. *Conjecture 6 with $\ell = 4$ and $t = k+1$ implies Conjecture 1.*

Proof. Assume Conjecture 6 holds for $\ell = 4$ and $t = k+1$. Suppose there is a k-blocked set P of at least n points. By Lemma 3, at most three points are collinear in P. Thus, the first conclusion of Conjecture 6 does not hold. By definition, the visibility graph of P is k-colorable. Thus, the second conclusion of Conjecture 6 does not hold. This contradiction proves that every k-blocked set has fewer than n points, and Conjecture 1 holds. □

Thus, since Conjecture 5 holds for $t \leq 5$, Conjecture 1 holds for $k \leq 4$. In Sect. 4 we take this result much further, by characterizing all representable 4-sets, thus concluding a tight bound on the size of a 4-blocked set.

Part of our interest in blocked point sets comes from the following.

Proposition 8. *For all $k \geq 3$ and $n \geq 2$, the edge set of the k-partite Turán graph $K(n,n,\ldots,n)$ can be partitioned into a set of "lines," where*

- *Each line is either an edge or an induced path on three vertices;*
- *Every pair of vertices is in exactly one line; and*
- *For every line L, there is a vertex adjacent to each vertex in L.*

Proof. Let (i,p) be the pth vertex in the ith color class for $i \in \mathbb{Z}_k$ and $p \in \mathbb{Z}_n$ (taken as additive cyclic groups). We introduce three types of lines. First, for $i \in \mathbb{Z}_k$ and distinct $p,q \in \mathbb{Z}_n$, let the triple $\{(i,p),(i+1,p+q),(i,q)\}$ be a line. Second, for $i \in \mathbb{Z}_k$ and $p \in \mathbb{Z}_n$, let the pair $\{(i,p),(i+1,p+p)\}$ be a line. Third, for $p,q \in \mathbb{Z}_n$ and distinct nonconsecutive $i,j \in \mathbb{Z}_k$, let the pair $\{(i,p),(j,q)\}$ be a line. By construction, each line is either an edge or an induced path on three vertices.

Every pair of vertices in the same color class are in exactly one line (of the first type). Consider vertices (i,p) and (j,q) in distinct color classes. First, suppose that i and j are consecutive. Without loss of generality, $j = i+1$. If $q \neq p+p$, then (i,p) and (j,q) are in exactly one line (of the first type). If $q = p+p$, then (i,p) and (j,q) are in exactly one line (of the second type). If i and j are not consecutive, then (i,p) and (j,q) are in exactly one line (of the third type). This proves that every pair of vertices is in exactly one line. Moreover, every edge of $K(n,n,\dots,n)$ is in exactly one line, and the lines partition the edges set.

Since every line L is contained in the union of two color classes, each vertex in neither color class intersecting L is adjacent to each vertex in L. $\square$

Proposition 8 is significant for Conjecture 5 because it says that $K(n,\dots,n)$ behaves like a "visibility space" with no four collinear points, but it has no large clique (for fixed k). Conjecture 1, if true, implies that such a visibility space is not "geometrically representable" for large n.

Let $b(n)$ be the minimum integer such that some set of n points in the plane in general position is blocked by some set of $b(n)$ points. Matoušek [6] proved that $b(n) \geq 2n-3$. Dumitrescu et al. [2] improved this bound to $b(n) \geq (\frac{25}{8} - o(1))n$. Many authors have conjectured or stated as an open problem that $b(n)$ is superlinear.

Conjecture 9 ([2, 6, 8, 9]). $\frac{b(n)}{n} \to \infty$ as $n \to \infty$.

Pór and Wood [9] proved that Conjecture 9 implies Conjecture 6, and thus implies Conjecture 1. That Conjecture 1 is implied by a number of other well-known conjectures, yet remains challenging, adds to its interest.

3 *k*-Blocked Sets with Small Color Classes

We now describe some methods for building blocked point sets from smaller blocked point sets.

Lemma 10. *Let G be a visibility graph. Let $i \in \{1,2,3\}$. Furthermore, suppose that if $i \geq 2$, then $V(G) \neq \emptyset$, and if $i = 3$, then not all the vertices of G are collinear. Let G_i be the graph obtained from G by adding an independent set of i new vertices, each adjacent to every vertex in G. Then G_1, G_2, and G_3 are visibility graphs.*

Proof. For distinct points p and q, let $\overleftarrow{pq}$ denote the ray that is (1) contained in the line through p and q, (2) starting at p, and (3) not containing q. Let $\mathcal{L}$ be the union of the set of lines containing at least two vertices in G.

$i = 1$: Since $\mathcal{L}$ is the union of finitely many lines, there is a point $p \notin \mathcal{L}$. Thus, p is visible from every vertex of G. By adding a new vertex at p, we obtain a representation of G_1 as a visibility graph.

$i = 2$: Let p be a point not in $\mathcal{L}$. Let v be a vertex of G. Each line in $\mathcal{L}$ intersects $\overleftarrow{vp}$ in at most one point. Thus, $\overleftarrow{vp} \setminus \mathcal{L} \neq \emptyset$. Let q be a point in $\overleftarrow{vp} \setminus \mathcal{L}$. Thus, p and q are visible from every vertex of G, but p and q are blocked by v. By adding new vertices at p and q, we obtain a representation of G_2 as a visibility graph.

$i = 3$: Let u, v, w be noncollinear vertices in G. Let p be a point not in $\mathcal{L}$ and not in the convex hull of $\{u, v, w\}$. Without loss of generality, $\overline{uv} \cap \overline{pw} \neq \emptyset$. There are infinitely many pairs of points $q \in \overleftarrow{up}$ and $r \in \overleftarrow{vp}$ such that w blocks q and r. Thus, there are such q and r both not in $\mathcal{L}$. By construction, u blocks p and q, and v blocks p and r. By adding new vertices at p, q, and r, we obtain a representation of G_3 as a visibility graph. □

Since no (≥ 3)-blocked set is collinear, Lemma 10 implies

Corollary 11. *If $k \geq 4$ and $\{n_1, \dots, n_{k-1}\}$ is representable and $n_k \in \{1,2,3\}$, then $\{n_1, \dots, n_{k-1}, n_k\}$ is representable.*

The representable (≤ 3)-sets were characterized in Theorem 2. In each case, each color class has at most three vertices. Now we characterize the representable (≥ 4)-sets, assuming that each color class has at most three vertices.

Proposition 12. *$\{n_1, \dots, n_k\}$ is representable whenever $k \geq 4$ and each $n_i \leq 3$, except for $\{1,3,3,3\}$.*

Proof. Say the k-set $\{n_1, \dots, n_k\}$ *contains* the $(k-1)$-set $\{n_1, \dots, n_{i-1}, n_{i+1}, \dots, n_k\}$ for each $i \in [k]$. We proceed by induction on $k \geq 4$. For the base case, assume that $k = 4$. If $\{n_1, n_2, n_3, n_4\}$ contains $\{1,1,1\}$, $\{1,1,2\}$, $\{1,2,2\}$, or $\{2,2,2\}$, then $\{n_1, n_2, n_3, n_4\}$ is representable by Theorem 2 and Corollary 11. In each remaining case, the following table describes a construction, except for $\{1,3,3,3\}$, which is not representable, as proved in Lemma 18.

$\{1,1,1,x\}$	Contains $\{1,1,1\}$	$\{1,1,2,x\}$	Contains $\{1,1,2\}$
$\{1,1,3,3\}$	Figure 1 minus $\{r_1, g_3, r_3, g_1\}$	$\{1,2,2,x\}$	Contains $\{1,2,2\}$
$\{1,2,3,3\}$	Figure 1 minus $\{g_1, g_3, r_3\}$	$\{2,2,2,x\}$	Contains $\{2,2,2\}$
$\{2,2,3,3\}$	Figure 1 minus $\{g_3, r_3\}$	$\{2,3,3,3\}$	Figure 1 minus g_3
$\{3,3,3,3\}$	Figure 1		

Now assume that $k \geq 5$: Consider a k-tuple $\{n_1, \dots, n_k\}$ with each $n_i \leq 3$. Say $n_1 \leq \cdots \leq n_k$. If $\{n_1, \dots, n_{k-1}\} \neq \{1,3,3,3\}$, then by induction $\{n_1, \dots, n_{k-1}\}$ is representable, which implies that $\{n_1, \dots, n_k\}$ is representable by Corollary 11. Otherwise, $k = 5$ and $\{n_1, \dots, n_5\} = \{1,3,3,3,3\}$, which contains the representable 4-tuple $\{3,3,3,3\}$, implying $\{n_1, \dots, n_5\}$ is representable by Corollary 11. □

4 4-Blocked Point Sets

As we saw in Sect. 2, Conjecture 1 holds for $k \le 4$. In this section we study 4-blocked point sets in more detail. First, we derive explicit bounds on the size of 4-blocked sets from other results in the literature. Then, following a more detailed approach, we characterize all representable 4-sets, to conclude a tight bound on the size of 4-blocked sets.

Proposition 13. *Every* 4*-blocked set has less than* 2^{790} *points.*

Proof. Abel et al. [1] proved that every set of at least $\mathrm{ES}(\frac{(2\ell-1)^{\ell}-1}{2\ell-2})$ points in the plane contains ℓ collinear points or an empty convex pentagon, where $\mathrm{ES}(k)$ is the minimum integer such that every set of at least $\mathrm{ES}(k)$ points in general position in the plane contains k points in convex position. Let P be a 4-blocked set. The visibility graph of P is 4-colorable, and thus contains no empty convex pentagon. By Lemma 3, at most three points in P are collinear. Thus, $|P| \le \mathrm{ES}(400) - 1$ by the above result with $\ell = 4$. Tóth and Valtr [12] proved that $\mathrm{ES}(k) \le \binom{2k-5}{k-2} + 1$. Hence, $|P| \le \binom{795}{398} < 2^{790}$. □

Lemma 14. *If P is a blocked set of n points with m points in the largest color class, then $n \ge 3m - 3$ and $n \ge (\frac{33}{8} - o(1))m$.*

Proof. If S is the largest color class, then $P - S$ blocks S. By Lemma 4, S is in general position. By the results of Matoušek [6] and Dumitrescu et al. [2] mentioned in Sect. 2, $n - m \ge 2m - 3$ and $n - m \ge (\frac{25}{8} - o(1))m$. □

Proposition 15. *Every* 4*-blocked set has fewer than* 2^{578} *points.*

Proof. Let P be a 4-blocked set of n points. Let S be the largest color class in P. Let $m := |S| \ge \frac{n}{4}$. By Lemma 14, $n \ge (\frac{33}{8} - o(1))m \ge (\frac{33}{32} - o(1))n$. Thus, $o(n) \ge \frac{n}{32}$. Hence, n is bounded. A precise bound is obtained as follows. Dumitrescu et al. [2] proved that S needs at least $\frac{25m}{8} - \frac{25m}{2\ln m} - \frac{25}{8}$ blockers, which come from the other color classes. Thus, $n - m \ge \frac{25m}{8} - \frac{25m}{2\ln m} - \frac{25}{8}$, implying $n \ge \frac{33m}{8} - \frac{25m}{2\ln m} - \frac{25}{8}$. Since $\frac{n}{4} \le m \le n$, we have $\frac{25n}{2\ln n} + \frac{25}{8} \ge \frac{n}{32}$. It follows that $n \le 2^{578}$. □

The next result is the simplest known proof that every 4-blocked point set has bounded size.

Proposition 16. *Every* 4*-blocked set has at most* 36 *points.*

Proof. Let P be a 4-blocked set. Suppose that $|P| \ge 37$. Let S be the largest color class. Thus, $|S| \ge 10$. By Lemma 4, S is in general position. By a theorem of Harborth [4], some 5-point subset $K \subseteq S$ is the vertex-set of an empty convex pentagon $\mathrm{conv}(K)$. Let $T := P \cap (\mathrm{conv}(K) - K)$. Since $\mathrm{conv}(K)$ is empty with respect to S, each point in T is not in S. Thus, T is 3-blocked. K needs at least eight blockers (five blockers for the edges on the boundary of $\mathrm{conv}(K)$, and three blockers for the chords of $\mathrm{conv}(K)$). Thus, $|T| \ge 8$. But every 3-blocked set has at most six points, which is a contradiction. Hence, $|P| \le 36$. □

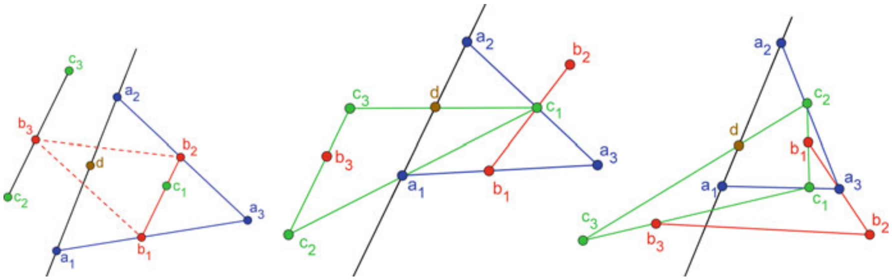


Fig. 3 Cases 1.1, 1.2, and 2 in Lemma 18

We now set out to characterize all representable 4-sets. We need a few technical lemmas.

Lemma 17. *Let A be a set of three monochromatic points in a 4-blocked set P. Then $P \cap \mathrm{conv}(A)$ contains a point from each color class.*

Proof. $P \cap \mathrm{conv}(A)$ contains at least three points in A. If $P \cap \mathrm{conv}(A)$ contains no point from one of the three other color classes, then $P \cap \mathrm{conv}(A)$ is a (≤ 3)-blocked set with three points in one color class (A), contradicting Theorem 2. □

Let $a * b * c$ mean that b blocks a and c, and let $a * b * c * d$ mean that $a * b * c$ and $b * c * d$. This allows us to record the order in which points occur on a line. Let $\overleftrightarrow{ab}$ be the line through two points a and b. For three noncollinear points a, b, c, let Δabc be the open triangle with vertices a, b, c.

Lemma 18. $\{3,3,3,1\}$ *is not representable.*

Proof. Suppose that P is a $\{3,3,3,1\}$-blocked set of points, with color classes $A = \{a_1, a_2, a_3\}$, $B = \{b_1, b_2, b_3\}$, $C = \{c_1, c_2, c_3\}$, and $D = \{d\}$.

If d does not block some monochromatic pair, then $P \setminus D$ is a 3-blocked set of nine points, contradicting Theorem 2. Therefore, d blocks some monochromatic pair, which we may call a_1, a_2. Now $\overleftrightarrow{a_1 a_2}$ divides the remaining seven points of P into two sets, P_1 and P_2, where without loss of generality, $4 \leq |P_1| \leq 7$ and $0 \leq |P_2| \leq 3$. If $|P_1| \geq 6$, then $P_1 \cup \{a_1\}$ is a 3-blocked set of more than six points, contradicting Theorem 2. Thus, $|P_1| = 4$ and $|P_2| = 3$, or $|P_1| = 5$ and $|P_2| = 2$. Consider the following cases, as illustrated in Fig. 3.

Case 1. $|P_1| = 4$ and $|P_2| = 3$.

By Theorem 2, P_1 is a $\{1,1,2\}$-blocked set. Thus, without loss of generality, $P_1 = \{a_3, b_1, b_2, c_1\}$ and $P_2 = \{b_3, c_2, c_3\}$, where $c_2 * b_3 * c_3$. Some point in P_1 blocks b_1 and b_2. If $b_1 * a_3 * b_2$, then neither b_1 nor b_2 can block $\overline{a_3 a_1}$ or $\overline{a_3 a_2}$; thus, c_1 blocks a_3 from both a_1 and a_2, a contradiction. Therefore, $b_1 * c_1 * b_2$. Since two of these three points must block $\overline{a_3 a_1}$ and $\overline{a_3 a_2}$, we may assume that $a_1 * b_1 * a_3$, and either $a_2 * b_2 * a_3$ or $a_2 * c_1 * a_3$.

Case 1.1. $a_2 * b_2 * a_3$: Since $c_2 * b_3 * c_3$, the only possible blockers for $\overline{b_1 b_3}$ are on $\overleftrightarrow{a_1 a_2}$. We cannot have $b_3 * a_1 * b_1$, for then $b_3 * a_1 * b_1 * a_3$. We cannot have $b_3 * a_2 * b_1$, for then $\overleftrightarrow{b_1 b_3}$ separates b_2 from d, implying $d \notin \text{conv}(B)$, contradicting Lemma 17. Therefore, $b_1 * d * b_3$. Similarly, $b_2 * d * b_3$, a contradiction.

Case 1.2. $a_2 * c_1 * a_3$: Thus, a_2 does not block $\overline{c_1 c_2}$ or $\overline{c_1 c_3}$. Therefore, a_1 and d block $\overline{c_1 c_2}$ and $\overline{c_1 c_3}$. Without loss of generality, $c_2 * a_1 * c_1$ and $c_3 * d * c_1$. Since $c_2 * b_3 * c_3$, the blocker for $\overline{b_1 b_3}$ is on the same side of $\overleftrightarrow{c_1 c_3}$ as a_1, and on the same side of $\overleftrightarrow{b_1 b_2}$ as a_1. The only such points are a_1, c_2, and b_3. Thus, a_1 or c_2 blocks $\overline{b_1 b_3}$. Now, c_2 does not block $\overline{b_1 b_3}$, as otherwise $b_1 * c_2 * b_3 * c_3$. Similarly, a_1 does not block $\overline{b_1 b_3}$, as otherwise $b_3 * a_1 * b_1 * a_3$. This contradiction concludes this case.

Case 2. $|P_1| = 5$ and $|P_2| = 2$.

We have $a_3 \in P_1$, as otherwise P_1 is 2-blocked, contradicting Theorem 2. Without loss of generality, $P_1 = \{a_3, b_1, b_2, c_1, c_2\}$ and $P_2 = \{b_3, c_3\}$. Note that a_3 must block at least one of $\overline{b_1 b_2}$ and $\overline{c_1 c_2}$, because $P_1 \setminus \{a_3\}$ is too large to be 2-blocked; however, it cannot block both, for then there would be no valid blockers left for a_3. Thus, without loss of generality, $b_1 * a_3 * b_2$ and $c_1 * b_1 * c_2$. Since $a_3 \in \overline{b_1 b_2}$, neither b_1 nor b_2 blocks $\overline{a_3 a_1}$ or $\overline{a_3 a_2}$. Thus, without loss of generality, $a_1 * c_1 * a_3$ and $a_2 * c_2 * a_3$.

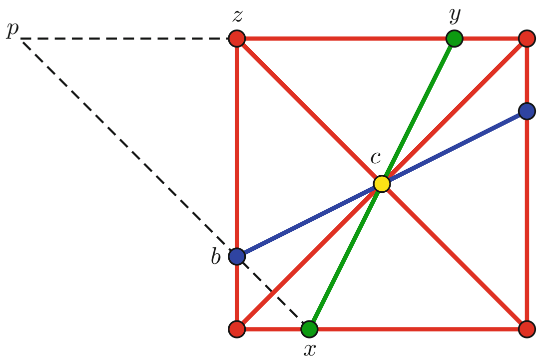
By Lemma 17, $P \cap \text{conv}\{c_1, c_2, c_3\}$ contains some member of A, say a_1. We cannot have $a_1 \in \overline{c_1 c_3}$, for then $c_3 * a_1 * c_1 * a_3$. Therefore, a_1 is on the same side of $\overleftrightarrow{c_1 c_3}$ as c_2; consequently, d and a_2 are, as well. It follows that $c_1 * b_3 * c_3$, and so b_3 sees b_2. □

Lemma 19. *Let P be a 4-blocked set. Suppose that some color class S of P contains a subset K, such that $|K| = 4$ and K is the vertex-set of a convex quadrilateral* $\text{conv}(K)$ *that is empty with respect to S. Then P is $\{4,2,2,1\}$-blocked.*

Proof. Let $T := P \cap (\text{conv}(K) - K)$. Since $\text{conv}(K)$ is empty with respect to S, each point in T is not in S. Thus, T is 3-blocked. K needs at least five blockers (four blockers for the edges on the boundary of $\text{conv}(K)$, and at least one blocker for the chords of $\text{conv}(K)$). The only representable 3-sets with at least five points are $\{2,2,1\}$ and $\{2,2,2\}$. At least four points in T are on the boundary of $\text{conv}(T)$. Every $\{2,2,2\}$-blocked set contains three points on the boundary of the convex hull. Thus, T is $\{2,2,1\}$-blocked. Hence, as illustrated in Fig. 4, one point c in T is at the intersection of the two chords of $\text{conv}(K)$, and exactly one point in T is on each edge of the boundary of $\text{conv}(K)$, such that the points on opposite edges of $\text{conv}(K)$ are collinear with c.

We claim that no other point is in P. Suppose otherwise, and let p be a point in P outside $\text{conv}(K)$ at a minimum distance from $\text{conv}(K)$. Let x be a point in $\text{conv}(K)$ receiving the same color as p. Thus, p and x are blocked by some point b in $\text{conv}(K)$. Hence, b and x are collinear with no other point in $P \cap \text{conv}(K)$, so $x \neq c$. Since p sees the nearest vertex in $\text{conv}(K)$, we also have $x \notin K$. Thus, x is in the interior of one of the edges of the boundary of $\text{conv}(K)$. Let y be the point in $\text{conv}(K)$ receiving the same color as x. Thus, x and y are on opposite edges of the boundary of $\text{conv}(K)$. Hence, p and y receive the same color, implying p and y are blocked. Since p is at a

Fig. 4 A $\{4,2,2,1\}$-blocked point set



minimum distance from $\mathrm{conv}(K)$, this blocker must be a point z of $\mathrm{conv}(K)$. Having both $p*b*x$ and $p*z*y$ implies that p,z,y and one other point of $\mathrm{conv}(K)$ are four collinear points, which contradicts Lemma 3. Hence, no other point is in P, and P is $\{4,2,2,1\}$-blocked. □

Lemma 19 has the following corollary (let $K := S$).

Corollary 20. *Let P be a 4-blocked set. Suppose that some color class S consists of exactly four points in convex position. Then P is $\{4,2,2,1\}$-blocked.*

The next lemma is a key step in our characterization of representable 4-sets.

Lemma 21. *Each color class in a 4-blocked point set has at most four points.*

Proof. Suppose that some 4-blocked point set P has a color class S with at least five points. Esther Klein [3] proved that every set of at least five points in general position in the plane contains an empty quadrilateral. By Lemma 4, S is in general position. Thus, S contains a subset K, such that $|K| = 4$ and K is the vertex set of a convex quadrilateral $\mathrm{conv}(K)$ that is empty with respect to S. By Lemma 19, P is $\{4,2,2,1\}$-blocked, which is the desired contradiction. □

Lemma 22. *Let P be a 4-blocked point set with color classes A,B,C,D. Suppose that no color class consists of exactly four points in convex position (that is, Corollary 20 is not applicable). Furthermore, suppose that some color class A consists of exactly four points in nonconvex position. Then P is $\{4,2,2,2\}$-blocked, as in Fig. 5 for example.*

Proof. By Lemma 21, each color class has at most four points. By assumption, every 4-point color class is in nonconvex position. We may assume that A is minimal in the sense that no other 4-point color class is within $\mathrm{conv}(A)$.

Let $Q := P \cap \mathrm{conv}(A)$. Thus, Q is 4-blocked, and one color class is A. By the minimality of A, each other color class in Q has at most three points. We first prove that Q is $\{4,2,2,2\}$-blocked, and then show that this implies that $P = Q$.

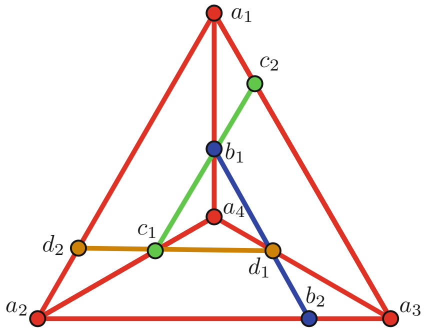


Fig. 5 A $\{4,2,2,2\}$-blocked point set

Let $A = \{a_1, a_2, a_3, a_4\}$, where a_4 is the interior point of $\mathrm{conv}(A)$. Note that the edges with endpoints in A divide $\mathrm{conv}(A)$ into three triangles with disjoint interiors. By Lemma 17, each color class of Q is represented in each of these triangles; this requires at least two points of each color (one of which could sit on the edge shared by two triangles).

We name a point with reference to its color class, such as $b_1 \in B$; or we name a point with reference to its position.

Let $Q' := Q \setminus A$. Let x_{ij} be the unique member of Q' that blocks $\overline{a_i a_j}$, for $1 \leq i < j \leq 4$. This accounts for exactly six points of Q'. For each case below we prove that Q' contains four mutually visible points (with respect to Q), which is a contradiction since Q' is 3-colored.

Q is not $\{4,3,3,3\}$-blocked, as otherwise we could delete the three outer members of A to represent $\{3,3,3,1\}$, which contradicts Lemma 18. Thus, Q is $\{4,3,2,2\}$-blocked, $\{4,3,3,2\}$-blocked, or $\{4,2,2,2\}$-blocked.

First, suppose that Q is $\{4,3,2,2\}$-blocked. Then Q' consists of six points x_{ij} and one additional point, y. The following three cases arise, as illustrated in Fig. 6.

Case 1. y blocks two points x_{i4} and x_{j4}: Without loss of generality, $x_{24} * y * x_{34}$. Now y is on one side or the other of $\overleftrightarrow{x_{14}a_4}$, and therefore y sees x_{12} or x_{13}. Without loss of generality, y sees x_{12}. Thus, x_{12}, x_{14}, x_{24}, and y are mutually visible points in Q'.

Case 2. y is collinear with, but does not block, two points x_{i4} and x_{j4}. Without loss of generality, $y * x_{24} * x_{34}$. Now x_{12} is on one side or the other of $\overleftrightarrow{yx_{24}}$; thus, x_{12} sees some $p \in \{x_{14}, x_{23}\}$, which is mutually visible with x_{12}, x_{24} and y; thus, we have four mutually visible points in Q'.

Case 3. y is not collinear with two points x_{i4} and x_{j4}: Then y sees all such points; thus, x_{14}, x_{24}, x_{34}, and y are mutually visible in Q'.

Now suppose that Q is $\{4,3,3,2\}$-blocked. Then Q' consists of the six points x_{ij} and two additional points, y_1 and y_2. Let $\delta = \{x_{14}, x_{24}, x_{34}\}$ and let δ_2 be the set of segments with both endpoints in δ. How many of the points y_i block members of δ_2? The following cases arise.

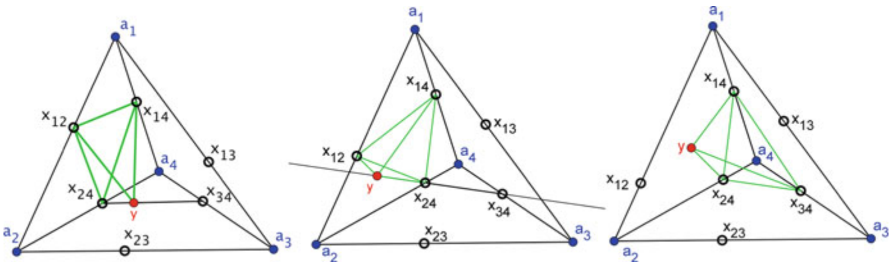
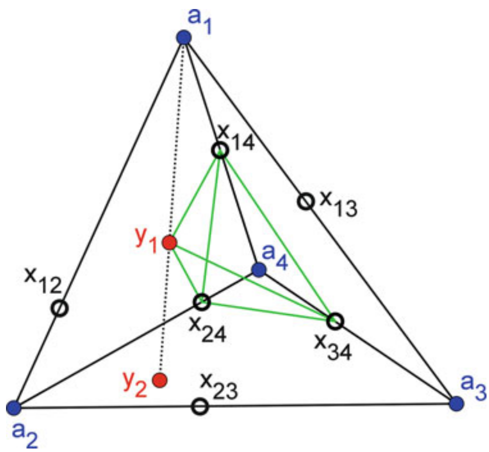


Fig. 6 Cases 1–3 when Q is $\{4,3,2,2\}$-blocked

Fig. 7 Case 1 when Q is $\{4,3,3,2\}$-blocked



Case 1. Neither y_1 nor y_2 blocks a segment in δ_2 (see Fig. 7).

Since y_1 and y_2 cannot block each other, without loss of generality, y_2 does not block y_1. Thus, δ and y_1 give four mutually visible points in Q'.

Case 2. At least one of y_1 and y_2 blocks a segment in δ_2.

Without loss of generality, y_1 blocks a segment in δ_2 and $x_{24} * y_1 * x_{34}$. Neither a_1 nor a_4 belongs to $\overleftrightarrow{x_{14}y_1}$, because otherwise *both* do (since $a_1 * x_{14} * a_4$), yielding four collinear points. Since $y_1 \in \Delta a_2a_3a_4 \subset \Delta a_2a_3x_{14}$, we know that a_2 and a_3 are on opposite sides of $\overleftrightarrow{x_{14}y_1}$ by the Crossbar Theorem. Thus, without loss of generality, a_4 is on the same side of $\overleftrightarrow{x_{14}y_1}$ as x_{24}—call this side "west" and the other side "east."

We now show that $\{x_{13}, x_{14}, x_{34}, y_1\}$ is a set of four points in general position. If not, then three of these points are collinear. Observe that x_{13}, x_{14}, and x_{34} are noncollinear. Thus, y_1 is one of the three collinear points. Since $\overleftrightarrow{x_{34}y_1}$ already contains a third point, namely x_{24}, it contains neither x_{13} nor x_{14} by Lemma 3. Thus, our three collinear points are $\{x_{13}, x_{14}, y_1\}$. Since $x_{24} * y_1 * x_{34}$, the point x_{34} is east of $\overleftrightarrow{x_{14}y_1}$. Thus, a_1 and a_3 are also east of $\overleftrightarrow{x_{14}y_1}$, so x_{13} must be as well. Therefore, x_{13} is not collinear with x_{14} and y_1. Thus, $\{x_{13}, x_{14}, x_{34}, y_1\}$ is a set of four points in general position.

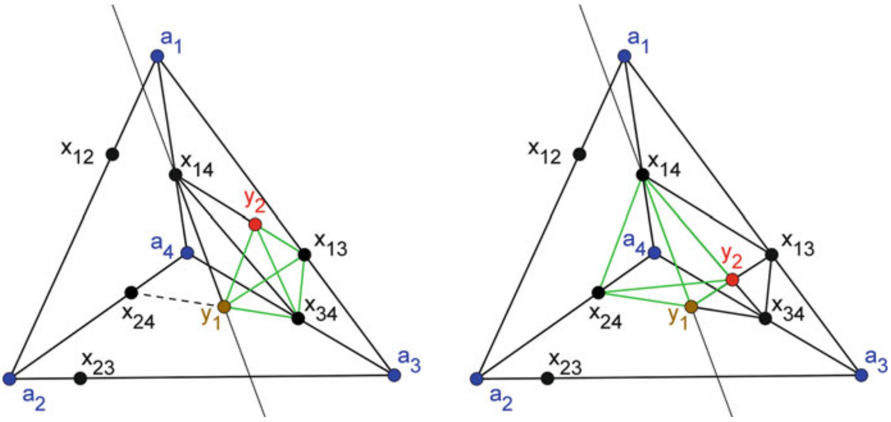


Fig. 8 Case 2a when Q is $\{4,3,3,2\}$-blocked

Since Q' does not contain four mutually visible points, there is some blocker in the midst of $\{x_{13}, x_{14}, x_{34}, y_1\}$. Because a_4 and x_{24} are west of $\overleftrightarrow{x_{14}y_1}$, the only possible blocker—which must be an interior point of $\mathrm{conv}(Q)$—is y_2, which must therefore belong to $\mathrm{conv}(x_{13}, x_{14}, x_{34}, y_1)$.

Case 2a. $\{x_{13}, x_{14}, x_{34}, y_1\}$ is in convex position (see Fig. 8).

If y_1 and x_{13} are on the same side of $\overleftrightarrow{x_{14}x_{34}}$, then y_1 and x_{24} would be on opposite sides of $\overleftrightarrow{x_{14}x_{34}}$, implying y_1 would not block $\overline{x_{24}x_{34}}$, which is a contradiction. Thus, y_1 and x_{13} are on opposite sides of $\overleftrightarrow{x_{14}x_{34}}$. It follows that $\mathrm{conv}(x_{13}, x_{14}, x_{34}, y_1)$ has chords $\overline{x_{14}x_{34}}$ and $\overline{x_{13}y_1}$. If y_2 blocks just one edge between $\{x_{13}, x_{14}, x_{34}, y_1\}$ (as in Fig. 8, left), then we may substitute y_2 for one endpoint of that edge to build a set of four mutually visible points in Q'. Thus, y_2 must block two edges between $\{x_{13}, x_{14}, x_{34}, y_1\}$ (as in Fig. 8, right). Hence, $x_{14} * y_2 * x_{34}$ and $y_1 * y_2 * x_{13}$. Now (1) a_4 is blocked by both x_{14} and x_{24}, and therefore blocks neither of them; (2) a_4 cannot block $\overline{y_1y_2}$, which is on the other side of $\overleftrightarrow{x_{14}y_1}$; (3) x_{34} is blocked by both y_1 and y_2, and therefore cannot block either of them; (4) x_{34} cannot block $\overline{x_{14}x_{24}}$, which is on the other side of $\overleftrightarrow{x_{14}y_1}$. Thus, $\{x_{14}, x_{24}, y_1, y_2\}$ is a set of four mutually visible points in Q'.

Case 2b. $\{x_{13}, x_{14}, x_{34}, y_1\}$ is not in convex position (see Fig. 9).

One of these four points lies inside the triangle formed by the other three. This interior point cannot be x_{13}, since it is on the boundary of $\mathrm{conv}(A)$. Nor can it be x_{14} or y_1, as we previously showed that x_{13} and x_{34} are on the same side (east) of $\overleftrightarrow{x_{14}y_1}$. Thus, x_{34} is in $\Delta x_{13}x_{14}y_1$. Hence, x_{13} and x_{14} are on opposite sides of $\overleftrightarrow{x_{34}y_1}$. That is, x_{13} and x_{23} are on the same side of $\overleftrightarrow{x_{34}y_1}$. This implies that x_{24} and y_1 are boundary points of $\mathrm{conv}(x_{23}, x_{24}, y_1, x_{13})$. Moreover, x_{13} and x_{23} must also be boundary points of $\mathrm{conv}(x_{23}, x_{24}, y_1, x_{13})$, as they are boundary points of $\mathrm{conv}(A)$.

Now $\{x_{23}, x_{24}, y_1, x_{13}\}$ is a set of four points in convex position. Note that $\mathrm{conv}(x_{13}, x_{14}, x_{34}, y_1)$ meets $\mathrm{conv}(x_{23}, x_{24}, y_1, x_{13})$ along only a single edge, $\overline{y_1x_{13}}$.

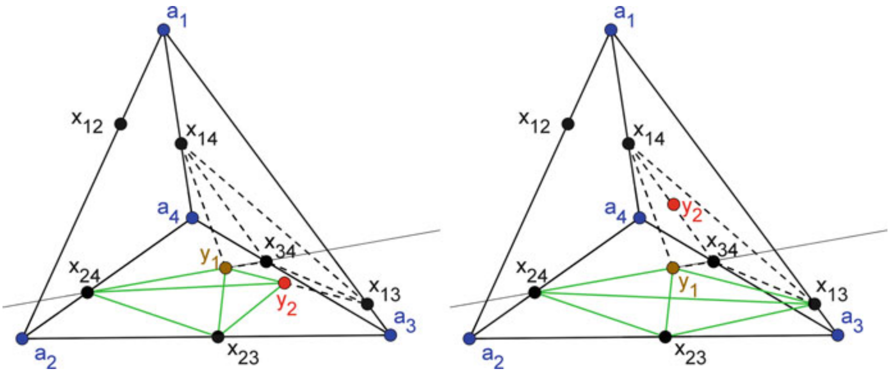


Fig. 9 Case 2b when Q is $\{4,3,3,2\}$-blocked

If y_2 blocks $\overline{y_1x_{13}}$, then $\{x_{23},x_{24},y_1,y_2\}$ is a set of four mutually visible points in Q' (as in Fig. 9, left). If y_2 does not block $\overline{y_1x_{13}}$, then $\{x_{23},x_{24},y_1,x_{13}\}$ is a set of four mutually visible points in Q' (as in Fig. 9, right).

The only remaining case is that Q is $\{4,2,2,2\}$-blocked. This is possible, as illustrated in Fig. 5. We now show that this point set is essentially the only $\{4,2,2,2\}$-blocked set, up to betweenness-preserving deformations. There are exactly enough points in color classes B, C, and D to block all edges between points in A. As each of the three A-triangles with point a_4 must contain a representative from each of the other three color classes, it follows that of the six A-edges, B blocks one interior edge and the opposite boundary edge, and likewise for C and D. Without loss of generality $b_1 = x_{14}$, $b_2 = x_{23}$, $c_1 = x_{24}$, $c_2 = x_{13}$, $d_1 = x_{34}$, and $d_2 = x_{12}$. Since b_1 blocks a_4, a_4 cannot block b_1. Since $\overline{b_1b_2}$ can be blocked only by an interior point of conv(A), it follows that either c_1 or d_1 blocks $\overline{b_1b_2}$. As these cases are symmetric, we may choose $c_1 \in \overline{b_1b_2}$. Now b_1 cannot block c_1, so c_1 must be blocked by d_1, which must in turn be blocked by b_1.

Now we show that $P = Q$. (This basically says that the point set in Fig. 5 cannot be extended without introducing a new color.) Suppose to the contrary that $P \setminus \text{conv}(A) \neq \emptyset$. Let x be a point in $P \setminus \text{conv}(A)$ closest to conv(A). Thus, x sees a vertex of conv(A), and $x \notin A$. Without loss of generality, $x \in B$. Recall that b_2 is on the line $\overleftrightarrow{a_2a_3}$. Thus, x is in the same half-plane determined by $\overleftrightarrow{a_2a_3}$ as the rest of Q, as otherwise b_2 would see $x \in B$. Which point blocks $\overline{b_1x}$? Not a_2 or a_3, for this would put x in the wrong half-plane. Not a_1, a_4, c_1, or c_2, since b_1 blocks each of these. Not d_1, which is on $\overline{b_1b_2}$. Therefore, $\overline{b_1x}$ can be blocked only by d_2. Since $d_1 \in \overline{b_1b_2}$ and $d_2 \in \overline{b_1x}$, it follows that b_2 and x (and any blocker between them) are on the same side of $\overleftrightarrow{d_1d_2}$. But the only other points of Q in that open half-plane are a_2 and a_3, which cannot block b_2. Thus, x sees b_2, which is a contradiction. Thus, $P = Q$ and P is $\{4,2,2,2\}$-blocked. $\square$

We now prove the main theorem of this section.

Theorem 23. *A 4-set $\{a,b,c,d\}$ is representable if and only if*

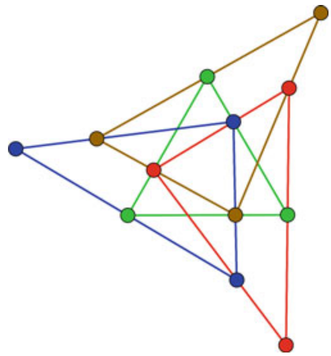


Fig. 10 Another $\{3,3,3,3\}$-blocked point set

- $\{a,b,c,d\} = \{4,2,2,1\}$*, or*
- $\{a,b,c,d\} = \{4,2,2,2\}$*, or*
- *All of* $a,b,c,d \leq 3$ *except for* $\{3,3,3,1\}$.

Proof. Figures 4 and 5 respectively show $\{4,2,2,1\}$-blocked and $\{4,2,2,2\}$-blocked point sets. When $a,b,c,d \leq 3$, the required constructions are described in Proposition 12. Now we prove that these are the only representable 4-sets. Let P be a 4-blocked point set. By Lemma 21, each color class has at most four points. Let S be the largest color class. If $|S| \leq 3$, then we are done by Proposition 12. Now assume that $|S| = 4$. If S is in nonconvex position, then P is $\{4,2,2,2\}$-blocked by Lemma 22. If S is in convex position, then P is $\{4,2,2,1\}$-blocked by Corollary 20. □

Corollary 24. *Every 4-blocked set has at most 12 points, and there is a 4-blocked set with 12 points.*

Note that in addition to the $\{3,3,3,3\}$-blocked set shown in Fig. 1, there is a different $\{3,3,3,3\}$-blocked point set, as illustrated in Fig. 10.

5 Midpoint-Blocked Point Sets

A k-blocked point set P is *k-midpoint-blocked* if for each monochromatic pair of distinct points $v,w \in P$ the midpoint of $\overline{vw}$ is in P. Of course, the midpoint of $\overline{vw}$ blocks v and w. A point set P is $\{n_1,\dots,n_k\}$*-midpoint-blocked* if it is $\{n_1,\dots,n_k\}$-blocked and k-midpoint-blocked. For example, the point set in Fig. 1 is $\{3,3,3,3\}$-midpoint-blocked.

Another interesting example is the projection[1] of $[3]^d$. With $d = 1$ this point set is $\{2,1\}$-blocked, with $d = 2$ it is $\{4,2,2,1\}$-blocked, and with $d = 3$ it is

[1]If G is the visibility graph of some point set $P \subseteq \mathbb{R}^d$, then G is the visibility graph of some projection of P to $\mathbb{R}^2$ (since a random projection of P to $\mathbb{R}^2$ is occlusion-free with probability 1).

$\{8,4,4,4,2,2,2,1\}$-blocked. In general, each set of points with exactly the same set of coordinates equal to 2 is a color class. Each color class of points with exactly i coordinates equal to 2 has size 2^{d-i}, and there are $\binom{d}{i}$ such color classes. Hence, $[3]^d$ is $\{\binom{d}{i} \times 2^{d-i} : i \in [0,d]\}$-midpoint-blocked and 2^d-midpoint-blocked.

We now prove Conjecture 1 when restricted to k-midpoint-blocked point sets. (Finally, we have weakened Conjecture 5 to something provable!)

A. Hernández-Barrera, F. Hurtado, J. Urrutia, and C. Zamora (On the midpoints of a plane point set, 2001, unpublished manuscript) introduced the following definition. Let $m(n)$ be the minimum number of midpoints determined by some set of n points in general position in the plane. Since the midpoint of $\overline{vw}$ blocks v and w, we have $b(n) \leq m(n)$. A. Hernández-Barrera, F. Hurtado, J. Urrutia, and C. Zamora (On the midpoints of a plane point set, 2001, unpublished manuscript) constructed a set of n points in general position in the plane that determine at most $cn^{\log 3}$ midpoints for some constant c. (All logarithms here are binary.) Thus, $b(n) \leq m(n) \leq cn^{\log 3} = cn^{1.585\cdots}$. This upper bound was improved independently by Pach [7] and Stanchescu [11] (and later by Matoušek [6]) to

$$b(n) \leq m(n) \leq nc^{\sqrt{\log n}}.$$

A. Hernández-Barrera, F. Hurtado, J. Urrutia, and C. Zamora (On the midpoints of a plane point set, 2001, unpublished manuscript) conjectured that $m(n)$ is superlinear (that is, $\frac{m(n)}{n} \to \infty$ as $n \to \infty$), which was verified by Pach [7]. More precise bounds were obtained by Stanchescu [11] and Sanders [10]. Applying the latest results on Freiman's Theorem, Pór and Wood [9] observed that for all $\varepsilon > 0$ there is an integer $N(\varepsilon)$ such that $m(n) \geq n(\log n)^{4/11-\varepsilon}$ for all $n \geq N(\varepsilon)$.

Theorem 25. *For each k, there is an integer n such that every k-midpoint-blocked set has at most n points. More precisely, for all $\varepsilon > 0$ and for all k, every k-midpoint-blocked set has at most $k \max\{N(\varepsilon), 2^{(k-1)^{11/(4-11\varepsilon)}}\}$ points.*

Proof. Let P be a k-midpoint-blocked set of n points. If $n \leq kN(\varepsilon)$, then we are done. Now assume that $\frac{n}{k} > N(\varepsilon)$. Let S be a set of exactly $s := \lceil \frac{n}{k} \rceil$ monochromatic points in P. Thus, S is in general position by Lemma 4. And for every pair of distinct points $v, w \in S$, the midpoint of $\overline{vw}$ is in $P - S$. Thus,

$$\tfrac{n}{k}(\log \tfrac{n}{k})^{4/11-\varepsilon} \leq m(s) \leq n - s \leq n(1 - \tfrac{1}{k}).$$

Hence, $(\log \frac{n}{k})^{4/11-\varepsilon} \leq k-1$, implying $n \leq k2^{(k-1)^{11/(4-11\varepsilon)}}$. The result follows. □

We now construct k-midpoint-blocked point sets with a "large" number of points. The method is based on the following product of point sets P and Q. For each point $v \in P \cup Q$, let (x_v, y_v) be the coordinates of v. Let $P \times Q$ be the point set $\{(v,w) : v \in P, w \in Q\}$, where (v,w) is at (x_v, y_v, x_w, y_w) in 4-dimensional space. For brevity we do not distinguish between a point in $\mathbb{R}^4$ and its image in an occlusion-free projection of the visibility graph of $P \times Q$ into $\mathbb{R}^2$.

Lemma 26. *If P is a $\{n_1,\dots,n_k\}$-midpoint-blocked point set and Q is a $\{m_1,\dots,m_\ell\}$-midpoint-blocked point set, then $P\times Q$ is $\{n_i m_j : i\in[k], j\in[\ell]\}$-midpoint-blocked.*

Proof. Color each point (v,w) in $P\times Q$ by the pair $(\mathrm{col}(v),\mathrm{col}(w))$. Thus, there are $n_i m_j$ points for the (i,j)th color class. Consider distinct points (v,w) and (a,b) in $P\times Q$.

Suppose that $\mathrm{col}(v,w)=\mathrm{col}(a,b)$. Thus, $\mathrm{col}(v)=\mathrm{col}(a)$ and $\mathrm{col}(w)=\mathrm{col}(b)$. Since P and Q are midpoint-blocked, $\frac{1}{2}(v+a)\in P$ and $\frac{1}{2}(w+b)\in Q$. Thus, $(\frac{1}{2}(v+a),\frac{1}{2}(w+b))$, which is positioned at $(\frac{1}{2}(x_v+x_a),\frac{1}{2}(y_v+y_a),\frac{1}{2}(x_w+x_b),\frac{1}{2}(y_w+y_b))$, is in $P\times Q$. This point is the midpoint of $\overline{(v,w)(a,b)}$. Thus, (v,w) and (a,b) are blocked by their midpoint in $P\times Q$.

Conversely, suppose that some point $(r,s)\in P\times Q$ blocks (v,w) and (a,b). Thus, $x_r=\alpha x_v+(1-\alpha)x_a$ for some $\alpha\in(0,1)$, $y_r=\beta y_v+(1-\beta)y_a$ for some $\beta\in(0,1)$, $x_s=\delta x_w+(1-\delta)x_b$ for some $\delta\in(0,1)$, and $y_s=\gamma y_w+(1-\gamma)y_b$ for some $\gamma\in(0,1)$. Hence, r blocks v and a in P, and s blocks w and b in Q. Thus, $\mathrm{col}(v)=\mathrm{col}(a)\neq\mathrm{col}(r)$ in P, and $\mathrm{col}(w)=\mathrm{col}(b)\neq\mathrm{col}(s)$ in Q, implying $(\mathrm{col}(v),\mathrm{col}(w))=(\mathrm{col}(a),\mathrm{col}(b))$.

We have shown that two points in $P\times Q$ are blocked if and only if they have the same color. Thus, $P\times Q$ is blocked. Since every blocker is a midpoint, $P\times Q$ is midpoint-blocked. □

Say P is a k-midpoint blocked set of n points. By Lemma 26, the i-fold product $P^i := P\times\cdots\times P$ is a k^i-blocked set of $n^i=(k^i)^{\log_k n}$ points. Taking P to be the $\{3,3,3,3\}$-midpoint-blocked point set in Fig. 1, we obtain the following result:

Theorem 27. *For all k a power of 4, there is a k-blocked set of $k^{\log_4 12}=k^{1.79\dots}$ points.*

This result describes the largest known construction of k-blocked or k-midpoint-blocked point sets. To promote further research, we make the following strong conjectures:

Conjecture 28. Every k-blocked point set has $O(k^2)$ points.

Conjecture 29. In every k-blocked point set there are at most k points in each color class.

Conjecture 29 would be tight for the projection of $[3]^d$ with $k=2^d$. Of course, Conjecture 29 implies Conjecture 28, which implies Conjecture 1.

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